$$\frac{\partial H}{\partial p} = \dot{x}$$

and

$$\frac{\partial H}{\partial x} = -\dot{p}.$$

Carrying out the partial derivatives, these become

$$\frac{\partial H}{\partial p} = \dot{x} = c$$

and

$$\frac{\partial H}{\partial x} = -\dot{p} = 0.$$

Thus, in the classical description of our particle, the momentum is conserved, and the position moves with fixed velocity *c*. In the quantum mechanical description, the whole probability distribution and the expectation value move with velocity *c*. In other words, the expectation value of position behaves according to the classical equations of motion.

## 9.2 NONRELATIVISTIC FREE PARTICLES

Only massless particles can move at the velocity of light, and I might add, they can only move at that velocity. All known particles other than photons and gravitons are massive and can move at any velocity less than *c*. When they move with a velocity *much* less than *c*, they are said to be nonrelativistic and their motion is governed by ordinary Newtonian